FUNDAMENTAL EQUATIONS OF MECHANICS

Kinematics

$$\boldsymbol{a} = \ddot{x}\,\hat{\boldsymbol{x}} + \ddot{y}\,\hat{\boldsymbol{y}} = \left(\ddot{\rho} - \rho\dot{\phi}^2\right)\hat{\boldsymbol{\rho}} + \left(\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}\right)\hat{\boldsymbol{\phi}}$$

Newton's Law

$$\sum \boldsymbol{F}_{\text{ext}} = \dot{\boldsymbol{p}} \qquad \sum \boldsymbol{\tau}_{\text{ext}} = \dot{\boldsymbol{L}}$$

Simple Rotational Motion

$$\boldsymbol{r}_{\text{cm}} = \frac{1}{M}\sum_{\alpha} m_\alpha \boldsymbol{r}_\alpha \qquad I = \sum_{\alpha} m_\alpha r_\alpha^2$$

$$\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F} \qquad \boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$$

Moments of Inertia

$$I_{\text{pt}} = I_{\text{ring}} = MR^2 \qquad I_{\text{rod ab end}} = \frac{1}{3}ML^2 \qquad I_{\text{rod ab mid}} = \frac{1}{12}ML^2$$

$$I_{\text{disk}} = \frac{1}{2}MR^2 \qquad I_{\text{solid sph}} = \frac{2}{5}MR^2 \qquad I_{\text{sph shell}} = \frac{2}{3}MR^2$$

Momentum and Energy

$$\boldsymbol{J} = \int \boldsymbol{F}\,dt \qquad \boldsymbol{J}_{\text{rot}} = \int \boldsymbol{\tau}\,dt$$

$$P = \frac{dE}{dt} \text{ or } P = \frac{dW}{dt} = Fv$$

Conservation Laws

$$\boldsymbol{p}_i + \sum \boldsymbol{J}_{\text{ext}} = \boldsymbol{p}_f \qquad \boldsymbol{L}_i + \sum \boldsymbol{J}_{\text{rot ext}} = \boldsymbol{L}_f$$

$$E_i + \sum W_{\text{ext}} + \sum Q_{ext} = E_f$$

FORCE AND ENERGY

	Force		**Energy**
	$\boldsymbol{F} = -\boldsymbol{\nabla}U$	$\leftrightarrow$	$\Delta U = -\int \boldsymbol{F}\cdot d\boldsymbol{r}$
Gravity (on earth)	$\boldsymbol{w} = m\boldsymbol{g}$		$U_g = mgh$
Gravity (in space)	$\boldsymbol{F}_g = -\frac{Gm_1m_2}{r^2}\hat{\boldsymbol{r}}$		$U_g = -\frac{Gm_1m_2}{r}$
Spring	$\boldsymbol{F}_{sp} = -kx\,\hat{\boldsymbol{x}}$		$U_{sp} = \frac{1}{2}kx^2$
External Force	$\boldsymbol{F}_{\text{ext}}$		$W = \int \boldsymbol{F}_{\text{ext}}\cdot d\boldsymbol{r}$
Kinetic Friction	$\boldsymbol{f}_k = -\mu_k N\,\hat{\boldsymbol{v}}_{\text{rel}}$		$W = \int \boldsymbol{f}_k\cdot d\boldsymbol{r}$
Drag	$\boldsymbol{D} = -(bv + cv^2)\,\hat{\boldsymbol{v}}$		$W = \int \boldsymbol{D}\cdot d\boldsymbol{r}$
Buoyancy	$\boldsymbol{F}_B = -m_f\boldsymbol{g}$		$U_B = -m_f gh$
Electric	$\boldsymbol{F}_{\text{el}} = q\boldsymbol{E}$		$U_{\text{el}} = qV$
Magnetic	$\boldsymbol{F}_{\text{mag}} = q\boldsymbol{v}\times\boldsymbol{B}$		$W = 0$
Magnetic	$\boldsymbol{F}_{\text{mag–wire}} = I\boldsymbol{\ell}\times\boldsymbol{B}$		$W = 0$
Normal Force	$0 \le N \le N_{\text{max}}$		$W = 0$
Tension	$0 \le T \le T_{\text{max}}$		$W = \int \boldsymbol{T}\cdot d\boldsymbol{r}$
Static Friction	$0 \le f_s \le \mu_s N$		$W = 0$
Kinetic Energy			$K = \frac{1}{2}mv^2$
Rotational Kinetic			$K_{\text{rot}} = \frac{1}{2}I\omega^2$
Chemical Energy			U_{chem}
Thermal Energy			U_{th}

OSCILLATIONS AND WAVES

$$f = \frac{1}{T} \qquad \omega = 2\pi f = \frac{2\pi}{T} \qquad k = \frac{2\pi}{\lambda} \qquad v = f\lambda$$

ODEs

$$\ddot{x} = +\omega_0^2 x \quad \longrightarrow \quad x(t) = A\cosh(\omega_0 t - \delta)$$
$$= B_1 e^{\omega_0 t} + B_2 e^{-\omega_0 t}$$
$$\ddot{x} = -\omega_0^2 x \quad \longrightarrow \quad x(t) = A\cos(\omega_0 t - \delta)$$

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0 \quad \longrightarrow$$

$$x_{\text{damped}}(t) = \begin{cases} \text{underdamped } \beta < \omega_0\text{:} & A_{\text{tr}} e^{-\beta t}\cos(\omega_1 t - \delta_{\text{tr}}) \quad \omega_1 = \sqrt{\omega_0^2 - \beta^2} \\ \text{overdamped } \beta > \omega_0\text{:} & B_{1_{\text{tr}}} e^{-\beta_- t} + B_{2_{\text{tr}}} e^{-\beta_+ t} \quad \beta_\pm = \beta \pm \sqrt{\beta^2 - \omega_0^2} \end{cases}$$

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0\cos\omega t \quad \longrightarrow \quad x_{\text{driven}}(t) = A\cos(\omega t - \delta) + x_{\text{damped}}(t)$$

$$A = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}} \quad \tan\delta = \frac{2\beta\omega}{\omega_0^2 - \omega^2} \quad Q = \frac{\omega_0}{2\beta}$$

COORDINATES

$$x = \rho\cos\phi \qquad \hat{\boldsymbol{x}} = \cos\phi\,\hat{\boldsymbol{\rho}} - \sin\phi\,\hat{\boldsymbol{\phi}}$$
$$y = \rho\sin\phi \qquad \hat{\boldsymbol{y}} = \sin\phi\,\hat{\boldsymbol{\rho}} + \cos\phi\,\hat{\boldsymbol{\phi}}$$

$$x = r\sin\theta\cos\phi \qquad \hat{\boldsymbol{x}} = \sin\theta\cos\phi\,\hat{\boldsymbol{r}} + \cos\theta\cos\phi\,\hat{\boldsymbol{\theta}} - \sin\phi\,\hat{\boldsymbol{\phi}}$$
$$y = r\sin\theta\sin\phi \qquad \hat{\boldsymbol{y}} = \sin\theta\sin\phi\,\hat{\boldsymbol{r}} + \cos\theta\sin\phi\,\hat{\boldsymbol{\theta}} + \cos\phi\,\hat{\boldsymbol{\phi}}$$
$$z = r\cos\theta \qquad \hat{\boldsymbol{z}} = \cos\theta\,\hat{\boldsymbol{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}$$

Power Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(a)(x-a)^n$$

DIFFERENTIAL OPERATORS

$$d\boldsymbol{\ell} = \hat{\boldsymbol{x}}\,dx + \hat{\boldsymbol{y}}\,dy + \hat{\boldsymbol{z}}\,dz$$
$$= \hat{\boldsymbol{\rho}}\,d\rho + \hat{\boldsymbol{\phi}}\,\rho d\phi + \hat{\boldsymbol{z}}\,dz$$
$$= \hat{\boldsymbol{r}}\,dr + \hat{\boldsymbol{\theta}}\,r d\theta + \hat{\boldsymbol{\phi}}\,r\sin\theta\,d\phi$$

$$\boldsymbol{\nabla} f = \hat{\boldsymbol{x}}\frac{\partial f}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial f}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial f}{\partial z}$$
$$= \hat{\boldsymbol{\rho}}\frac{\partial f}{\partial \rho} + \hat{\boldsymbol{\phi}}\frac{1}{\rho}\frac{\partial f}{\partial \phi} + \hat{\boldsymbol{z}}\frac{\partial f}{\partial z}$$
$$= \hat{\boldsymbol{r}}\frac{\partial f}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\boldsymbol{\phi}}\frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$
$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$
$$= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\,A_\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial \phi}$$

$$\boldsymbol{\nabla}\times\boldsymbol{A} = \hat{\boldsymbol{x}}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{\boldsymbol{y}}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{\boldsymbol{z}}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$
$$= \hat{\boldsymbol{\rho}}\left(\frac{1}{\rho}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right) + \hat{\boldsymbol{\phi}}\left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right) + \hat{\boldsymbol{z}}\frac{1}{\rho}\left(\frac{\partial}{\partial \rho}(\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi}\right)$$
$$= \hat{\boldsymbol{r}}\frac{1}{r\sin\theta}\left(\frac{\partial}{\partial\theta}(\sin\theta\,A_\phi) - \frac{\partial A_\theta}{\partial\phi}\right) + \hat{\boldsymbol{\theta}}\left(\frac{1}{r\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{1}{r}\frac{\partial}{\partial r}(rA_\phi)\right) + \hat{\boldsymbol{\phi}}\frac{1}{r}\left(\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right)$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$
$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial f}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial\phi^2} + \frac{\partial^2 f}{\partial z^2}$$
$$= \frac{1}{r}\frac{\partial^2}{\partial r^2}(rf) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}$$

LAGRANGIAN MECHANICS

Calculus of Variations

$$J = \int_{x_1}^{x_2} f[y(x), y'(x), x]\, dx \qquad \frac{\partial f}{\partial y} = \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)$$

Lagrange's Equations

$$\mathcal{L} = K - U$$

$$S = \int_{t_1}^{t_2} \mathcal{L}\, dt \qquad \frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right)$$

CENTRAL FORCES

$$\mu\ddot{r} = -\frac{\partial U}{\partial r} + \frac{L^2}{\mu r^3} \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$L = \mu r^2 \dot{\phi} \qquad E = \frac{\gamma^2 \mu}{2L^2}(\epsilon^2 - 1)$$

$$F(r) = -\frac{\gamma}{r^2} \qquad U = -\frac{\gamma}{r}$$

Kepler Orbits

$$r(\phi) = \frac{c}{1 + \epsilon\cos\phi} \qquad c = \frac{L^2}{\mu\gamma} \qquad \epsilon = \frac{AL^2}{\mu\gamma}$$

$$\frac{(x+d)^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad a = \frac{c}{1-\epsilon^2} \qquad b = \frac{c}{\sqrt{1-\epsilon^2}} \qquad d = \epsilon a$$

SCATTERING THEORY

$$dN_{sc} = N_{inc} n_{tar} \frac{d\sigma}{d\Omega} d\Omega$$

$$\frac{d\sigma}{d\Omega} = \frac{b}{\sin\theta}\left|\frac{db}{d\theta}\right|$$

NON-INERTIAL REFERENCE FRAMES

$$\frac{d\boldsymbol{B}}{dt} = \boldsymbol{\omega} \times \boldsymbol{B} \qquad \boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{r} \qquad \boldsymbol{B}, \boldsymbol{r} \text{ fixed in rotating frame}$$

Fictitious Forces

Inertial:	$-m\boldsymbol{A}$
Centrifugal:	$-m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r})$
Coriolis:	$-2m\boldsymbol{\Omega} \times \boldsymbol{v}$
Azimuthal:	$-m\dot{\boldsymbol{\Omega}} \times \boldsymbol{r}$

RIGID-BODY DYNAMICS

Center of mass: $\boldsymbol{R} = \frac{1}{M}\sum_\alpha m_\alpha \boldsymbol{r}_\alpha \qquad \boldsymbol{r}_\alpha = \boldsymbol{R} + \boldsymbol{r}'_\alpha$

$$I_{ij} = \sum_\alpha m_\alpha \left(r_\alpha^2 \delta_{ij} - r_{\alpha_i} r_{\alpha_j}\right)$$

$$\boldsymbol{P} = M\dot{\boldsymbol{R}} \qquad \boldsymbol{L} = \boldsymbol{R} \times \boldsymbol{P} + \sum_\alpha m_\alpha \boldsymbol{r}'_\alpha \times \dot{\boldsymbol{r}}'_\alpha = \boldsymbol{R} \times \boldsymbol{P} + \boldsymbol{I} \cdot \boldsymbol{\omega}$$

$$\sum \boldsymbol{F}_{\text{ext}} = \dot{\boldsymbol{P}} \qquad \sum \boldsymbol{\tau}_{\text{ext}} = \dot{\boldsymbol{L}}$$

$$K = \frac{1}{2}M\dot{\boldsymbol{R}}^2 + \frac{1}{2}\sum_\alpha m_\alpha \dot{\boldsymbol{r}}'^2_\alpha = \frac{1}{2}M\dot{\boldsymbol{R}}^2 + \frac{1}{2}\boldsymbol{\omega} \cdot \boldsymbol{L}$$

Euler's Equations (body frame)

$$\lambda_1\dot{\omega}_1 - (\lambda_2 - \lambda_3)\omega_2\omega_3 = \tau_1$$

$$\lambda_2\dot{\omega}_2 - (\lambda_3 - \lambda_1)\omega_3\omega_1 = \tau_2$$

$$\lambda_3\dot{\omega}_3 - (\lambda_1 - \lambda_2)\omega_1\omega_2 = \tau_3$$